Visual Question Answering From Charts

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***Abstract*—Visual Question Answering (VQA) over charts is a challenging multimodal task that requires understanding both visual and textual content embedded within data visualizations. Traditional VQA systems, especially those built for natural images or structured tables, struggle when applied to bar charts, pie charts, and line plots, which involve spatial, color, and numeric reasoning. Previous approaches such as DVQA [3], FigureQA [4], and PlotQA [1] have laid the foundation by introducing datasets and synthetic benchmarks. However, many of these systems rely heavily on Optical Character Recognition (OCR), hand-crafted rules, or cannot generalize well to diverse real-world charts.**

In this paper, we propose a hybrid pipeline combining multiple object detection models and language understanding to answer questions about charts. Specifically, we integrate both YOLOv5 and Faster R-CNN to detect and localize chart elements such as bars, legends, and axis labels. The detected components are then converted into a structured table using annotated bounding boxes. This table is passed to transformer-based question answering models, such as T5 or UnifiedQA, which generate natural language answers.

Our experiments demonstrate that combining YOLOv5 and Faster R-CNN improves detection robustness across diverse chart layouts. Initial attempts using CNN-based models for visual analysis alone proved computationally expensive and less effective. The dataset comprises annotated chart images from HuggingFace and additional synthetic examples, ensuring a wide range of chart types and question formats. The entire pipeline was implemented and tested on Google Colab Pro using GPU acceleration. Compared to OCR-dependent pipelines and prior works like ChartQA [2], DePlot [6], and STL-CQA [?], our method offers faster performance and improved accuracy on visual and logical reasoning tasks.

The contributions of this work include: (1) integrating YOLOv5 and Faster R-CNN for accurate chart component detection, (2) leveraging pre-trained transformer QA models for advanced natural language understanding, and (3) presenting a robust, end-to-end system for real-world chart-based VQA.

***Index Terms*—Chart VQA, Faster R-CNN, OCR, Detectron2, COCO Dataset, Visual Reasoning.**

I. Introduction

In today's data-driven world, charts and visual plots are fundamental tools for communicating complex datasets efficiently. From scientific publications and financial dashboards to news articles and research reports, visualizations like bar charts, line graphs, and pie charts play a crucial role in making data interpretable. Despite their simplicity for human readers, extracting meaningful information from such visualizations remains a challenging task for machines. Visual Question Answering (VQA) on chart images is a growing field that combines computer vision, natural language understanding, and reasoning to answer natural language questions about data visualizations.

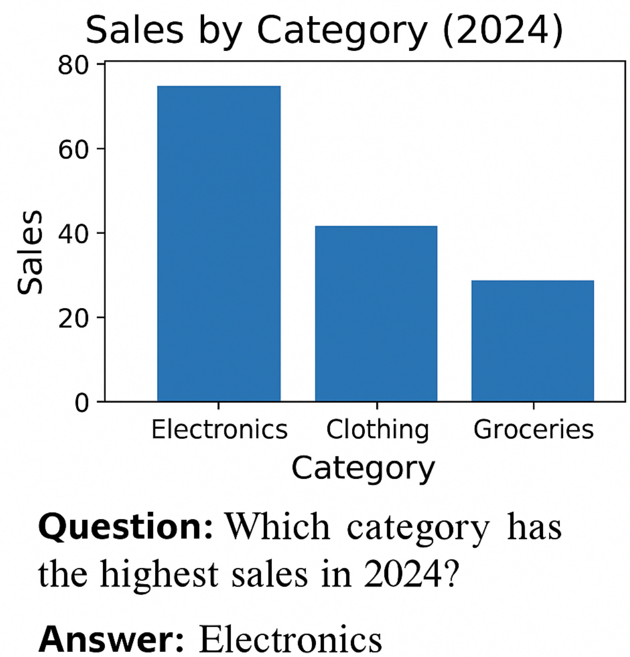

Fig. 1. Sample questions in our dataset

Traditional VQA systems have focused mainly on natural images, where the content is either visual or semantic but not both. However, charts are hybrid artifacts—they contain structured numerical and categorical data encoded visually using lines, bars, legends, colors, and textual labels. Therefore, a VQA system targeting charts must combine capabilities from both image understanding and tabular reasoning.

Several benchmark datasets and models have emerged to address this intersection. DVQA [3] introduced synthetic charts with programmatically generated questions, while FigureQA emphasized figure reasoning but used limited real-world semantics. PlotQA [1] advanced the field by providing large-scale real-world charts and diverse question types. ChartQA [2] further combined both visual and logical reasoning but still relied on Vision-TAPAS for semantic understanding, which struggled with highly visual tasks. Newer models such as DePlot [6] and MATCHA [7] employed one-shot plot-to-table translation and multimodal pretraining but required heavy compute resources and were sensitive to chart layout variance.

To address these limitations, we propose a hybrid approach that integrates a deep learning-based object detection model (YOLOv5) to detect chart elements (axes, bars, legends, titles, etc.) and a language model (T5 [21]) to understand and reason over the extracted chart data. Our method first uses bounding box annotations from YOLO to segment and extract chart entities, which are then flattened into a structured table format. This derived table is then passed to a T5-based model fine-tuned for question answering.

We initially explored a CNN-based pipeline similar to early vision architectures such as AlexNet and VGG , but the training overhead and slower inference made them less suitable for our real-time goals. YOLOv5 provided faster and more robust detection, allowing efficient component localization, which is crucial for real-world deployment.

Our system was developed and tested on Google Colab Pro with GPU support using datasets sourced from HuggingFace's version of ChartQA and DVQA. The system supports multiple chart types and question formats including arithmetic, color-based, logical, and visual reasoning. The results indicate our pipeline performs competitively against prior works and offers better speed and adaptability.

Through this work, we aim to bridge the gap between structured reasoning and visual understanding in charts, with future implications in automated report generation, data analysis, and multimodal search interfaces.

II. Literature Review

A. Dataset

Previous studies have emphasized advancing chart comprehension through datasets (shown in Table I) and models that address real-world complexity, such as handling out-of-vocabulary (OOV) answers, numerical reasoning, and multi-modal integration. [1], [2], and [3] prioritize large-scale, diverse question-answer pairs to challenge models with tasks like arithmetic operations and chart-specific terminology. These datasets share a focus on bridging synthetic and real-world scenarios, though their implementations differ: PlotQA [1] leverages real-world data sources (e.g., World Bank) for numerical OOV challenges, while ChartQA [2] balances human-written analytical questions with machine-generated templated queries to ensure breadth and depth. [1] presents an extensive collection of 28 million question–answer pairs extracted from 224K scientific plots. This dataset is designed to challenge models with numerical reasoning and out-of-vocabulary answers. In a similar way, [2] compiles 20,882 charts accompanied by over 32,000 questions—crafted both by humans and generated automatically—to support tasks requiring arithmetic and logical reasoning. Focusing on bar charts, [3] assembles more than 3 million image–question pairs, which emphasize the need to process chart-specific text. Adding further diversity, [3] utilizes data from 52 real-world charts along with hundreds of human-generated questions and visual explanations, thereby enriching the dataset with interpretability aspects. Meanwhile, they leverages the ChartSFT dataset to cover multiple chart types, supporting tasks such as summarization and numerical reasoning. Complementary to these, Deplot and matcha build upon datasets like PlotQA and ChartQA to convert visual plots into structured tables and incorporate mathematical reasoning into the learning process. For retrieval and summarization tasks, [15] adapts existing chart datasets, and [9] offers a benchmark consisting of 44,096 charts for generating natural language summaries. Finally, although primarily text-based, [10] has significantly influenced multimodal research by providing over 100,000 question–answer pairs that set a high standard for precise span extraction.

TABLE I
Comparison between existing datasets

Dataset	Size	Types
PlotQA	224,377 plots	Line, bar, scatter plots
ChartQA	20,882 charts	Bar, line, pie
DVQA	3M image–question pairs	Bar charts only
Chart-to-Text	44,096 charts	Bar, line, pie, radar, box plots

B. Data Extraction

Previous studies by PlotQA [1] cannot be considered as conclusive because it leverages a TableQA framework to tackle open-vocabulary questions, yet it neglects the integration of visual chart elements—such as color and layout which are important for visual reasoning. As the authors note earlier, more work is necessary to unify data tables and visual attributes for tasks like arithmetic operations. This gap is addressed by [2] which explicitly incorporates visual references into benchmark, enabling models to reason over both data tables and visual attributes for tasks like arithmetic operations. Many models rely on structured data extraction (e.g., OCR, table parsing) to translate visual charts into machine-readable formats. Hybrid models [1] and [6] exemplify this approach, combining OCR with table extraction for QA tasks. However, innovations diverge in their emphasis: [7] integrates math reasoning pretraining to enhance numerical accuracy, whereas chart paper adopts a multitask framework for zero-shot generalization across chart types. Transformer-based architectures like T5 and VisionTaPas [2] unify visual and textual features, while [3] DVQA employs dynamic encoding to handle chart-specific vocabularies.

A recurring challenge across studies is the loss of visual context (e.g., color, spatial relationships) during data extraction, which impacts performance on tasks requiring color-based reasoning or positional analysis. While [6] DePLOT achieves high accuracy in table reconstruction (97.1% RNSS), it struggles with color-dependent queries, a limitation also noted in MATCHA [7]. Similarly, Chart-to-Text [16] and Text-to-Text [21] models generate fluent summaries but face hallucinations and factual inaccuracies, reflecting broader issues in balancing fluency and precision.

In contrast, SQuAD [10], though text-focused, underscores the universal gap between human and machine comprehension, with machines achieving 51% F1 versus humans at 86.8%. This mirrors challenges in chart VQA [3], where even state-of-the-art models like DePLOT + Codex (67.6% accuracy on ChartQA) fall short of human-like robustness.

The papers collectively advocate for multimodal fusion and specialized pretraining as pathways forward. Yet, their strategies vary: some prioritize architectural innovations (e.g., MATCHA's [7] derendering), while others focus on expanding dataset diversity (e.g., Chart-to-Text's 44,096 charts [16]). These efforts highlight a shared vision of interpretable, scalable chart comprehension tools, tempered by the need to address persistent gaps in visual context preservation and high-precision reasoning. Papers that survey VQA more broadly 2024survey also highlight multiple datasets centered on tasks ranging from synthetic imagery to interactive 3D environments, illustrating diverse yet complementary pathways for advancing multimodal question answering research.

C. Model

The modeling approaches in these studies are equally varied. In [1], a hybrid model integrates a classification pipeline with a multi-stage process—combining visual element detection, OCR, and structured table extraction—to address both fixed and open-vocabulary queries. [2] employs transformer-based architectures VisionTaPas which is an extension of TaPas for QA over charts that blend visual features with extracted numerical data to perform arithmetic and logical operations. In [3], a dual-network design enhanced by dynamic encoding is introduced to effectively manage the unique text elements within bar charts. [4] proposes a pipeline that transforms visual queries into data-based ones and generates template-driven visual explanations, thus enhancing interpretability. [5] implements a two-stage training method that first converts charts into tables and then applies multitask instruction tuning, ensuring improved performance across diverse tasks. Further advancing these methods, [6] utilizes an end-to-end Transformer to convert plots into linearized tables, empowering one-shot reasoning by large language models, while [7] enriches visual language pretraining by incorporating explicit mathematical reasoning alongside chart derendering. Additional studies such as [8] and [9] integrate OCR-based extraction, vision-language models, and neural techniques inspired by image captioning to support effective chart retrieval and summarization. The foundational approaches described in [10] continue to influence these models by demonstrating the efficacy of precise text span extraction.

Underlying these innovations are challenges that persist across the various tasks. [1] highlights the difficulty in performing intricate numerical reasoning and detecting complex visual elements, particularly when responses extend beyond a fixed vocabulary. The approaches in [2] and [3] reveal that even with advanced transformer and dual-network models, operations that require nested computations and the extraction of nuanced, chart-specific text remain demanding. The system described in [4], while enhancing interpretability with visual explanations, still encounters issues with simpler query types and natural language fluency. [5] demonstrates that aligning graphical elements with implicit numerical data is a continuous hurdle, and while [6] achieves high accuracy in table reconstruction, some critical visual details—such as color—can be lost in translation. In [7], the integration of mathematical reasoning yields notable performance improvements, yet high-precision computations and the preservation of layout nuances continue to present challenges. Furthermore, the retrieval methods in [8] and the summarization strategies in [9] illustrate that incorporating multiple modalities can enhance performance, even as these approaches must navigate the complexity of diverse chart formats. Finally, the text comprehension framework established in [10] underscores the ongoing gap between machine and human understanding in the realm of precise information extraction.

Together, these works weave a comprehensive narrative that advances our understanding of how machines can interpret, analyze, and explain complex visual data. Each study—whether by introducing novel datasets, pioneering innovative models, or identifying critical challenges—adds a unique dimension to the evolving field of visual and language reasoning.

D. Challenges

Current research reveals several key challenges that persist in the domain of visual question answering and chart reasoning. In [1], models are challenged by the need to perform complex numerical reasoning while accurately detecting intricate visual elements, especially when confronted with out-of-vocabulary responses. Similarly, [2] identifies difficulties in handling nested operations and integrating arithmetic with logical reasoning, which remain problematic even for advanced transformer-based architectures. [3] further underscores the issue by demonstrating that processing chart-specific text in bar charts requires more robust and dynamic language modeling techniques. Moreover, the pipeline proposed in [4]—which focuses on generating visual explanations—still struggles with simpler query types and achieving natural language fluency in its output.

Enhancements are required across multiple dimensions to advance the state-of-the-art. [5] suggests that there is considerable room for improvement in aligning graphical elements with implicit numerical data, indicating a need for models that can seamlessly integrate multi-modal inputs without losing critical contextual information. Although [6] demonstrates impressive accuracy in converting plots to structured tables (with table reconstruction accuracies reaching 97.1% and 94.2% on specific metrics), the loss of vital visual attributes such as color and layout calls for enhanced methods that preserve these details. [7] takes an important step by incorporating mathematical reasoning into visual pretraining; however, high-precision computations and effective preservation of layout nuances remain challenging.

In the realms of retrieval and summarization, as explored in [8] and [9], integrating multiple modalities to produce factually correct and coherent natural language summaries is still an open problem. Furthermore, the text-based comprehension techniques illustrated by [10] indicate that even in comparatively simpler domains, a significant gap exists between machine and human performance.

Looking ahead, further enhancements are essential. Developing more integrated models that can concurrently handle visual, textual, and numerical data without compromising on any front is a key direction. Techniques that preserve crucial visual attributes—such as color, spatial layout, and contextual metadata—would greatly improve the fidelity of chart-to-table and chart-to-text translations. Additionally, incorporating advanced reasoning modules capable of nested, high-precision computations, perhaps through innovative pretraining paradigms or hybrid architectures, could help bridge the performance gap between current models and human-level understanding. These improvements are crucial for pushing the boundaries of what current visual reasoning systems can achieve, ultimately enabling more accurate and interpretable machine understanding of complex visual data.

III. Dataset

The dataset used in this study comprises visual chart samples annotated with their respective chart types. Each chart is represented by a filename and labeled as one of four types: `line`, `v_bar` (vertical bar), `h_bar` (horizontal bar), or `pie`. The data was split into training, validation, and testing subsets, with care taken to maintain consistent class distribution across all splits.

The complete dataset consists of **18,317** unique samples. The distribution across the splits is summarized in TableII.

TABLE II
Distribution of Chart Types Across Dataset Splits

Chart Type	Train Dataset	Val Dataset	Test Dataset
Vertical Bar	7110 (55.46%)	1524 (55.46%)	1524 (55.46%)
Horizontal Bar	3796 (29.61%)	814 (29.62%)	814 (29.62%)
Line	1536 (11.98%)	329 (11.97%)	329 (11.97%)
Pie	379 (2.96%)	81 (2.95%)	81 (2.95%)
Total Samples	**12821**	**2748**	**2748**

In addition to the chart type classification data, the dataset includes text annotations corresponding to each chart. The total number of annotated input records is as follows:

- Training set: 19,789 inputs
- Validation set: 4,276 inputs
- Test set: 4,234 inputs

Moreover, two additional JSON datasets were used to enrich the training data: `augmented.json` and `human.json`, consisting of 20,901 and 7,398 items respectively. Their distribution is provided in Table III.

TABLE III
Distribution of Augmented and Human JSON Samples

Source	Train	Validation	Test	Total
Augmented	14,591	3,208	3,102	20,901
Human	5,198	1,068	1,132	7,398
Total	**19,789**	**4,276**	**4,234**	**28,299**

This comprehensive and balanced dataset provides a strong foundation for training and evaluating models for automatic chart type classification and chart-to-text generation.

IV. Methodology

This section describes the components of our Visual Question Answering (VQA) system designed for chart images: the feature extraction models (YOLO and Faster R-CNN), and the T5-based question answering model. It also details data preprocessing, training setups, and how the inputs are prepared for effective learning.

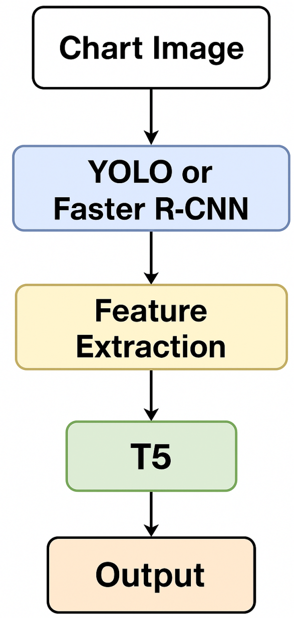

Fig. 2. Flow diagram illustrating the overall process of the proposed method

A. Feature Extraction from Chart Images

1) Object Detection Models: We utilize two object detection frameworks—YOLO and Faster R-CNN—to identify and localize key visual elements in chart images, such as bars, legends, axis labels, and data points.

a) YOLO (You Only Look Once): YOLO treats object detection as a single regression problem, dividing the input image into an $S \times S$ grid. Each grid cell predicts bounding boxes along with confidence scores and class probabilities for detected objects. This enables fast, real-time detection with reasonable accuracy. We employ YOLOv5 for its balance between speed and performance and fine-tune it on our annotated chart dataset. The model learns to recognize chart components and outputs bounding boxes and class labels, which serve as visual features for subsequent processing [18].

b) Faster R-CNN: Faster R-CNN is a two-stage detector that first uses a Region Proposal Network (RPN) to generate candidate object regions, followed by a classification and bounding box refinement stage. Due to its region-based approach, Faster R-CNN typically provides higher accuracy, especially for charts with complex or overlapping elements.

We fine-tune Faster R-CNN on the same dataset to improve detection precision, particularly for subtle or small chart components [19].

2) Text Information Extraction: Unlike conventional pipelines that depend on Optical Character Recognition (OCR) for extracting text from images, our method directly uses the ground-truth textual annotations available in the dataset. This eliminates errors related to OCR misreads and improves the reliability of textual features associated with detected objects. The bounding boxes detected by YOLO and Faster R-CNN are mapped to corresponding textual annotations, providing rich semantic context for each visual element.

B. Data Preprocessing and Training

1) Dataset Preparation: Our dataset consists of chart images annotated with bounding boxes for visual elements and associated textual labels. For training the detection models, images are resized to fixed dimensions and normalized. Bounding box coordinates and class labels are encoded according to the model's input format.

2) Training Setup:

a) YOLO: We initialize YOLOv5 with pretrained weights on the COCO dataset and fine-tune it for 50 epochs using the Adam optimizer with a learning rate. Data augmentation techniques, including random scaling and flipping, are applied to improve generalization.

b) Faster R-CNN: We fine-tune Faster R-CNN with a ResNet-50 backbone pretrained on ImageNet. Training is conducted for 30 epochs using stochastic gradient descent (SGD) with a learning rate and momentum of 0.9. Batch size is set to 16, and standard data augmentations are applied.

C. Question Answering with T5

1) Input Formatting: The question answering model is built upon the T5 transformer, which operates on text inputs and outputs. For each chart-question pair, we construct a linearized input sequence that concatenates:

- The question in natural language,
- The extracted table data from annotations (formatted as key-value pairs or tabular text),
- The textual labels corresponding to detected visual features.

This format allows T5 to jointly attend to both the question and the chart content encoded as text.

2) Model Training: We fine-tune the pretrained T5-base model on a dataset of chart-question-answer triples. The input is the linearized combination described above, and the target is the expected answer text.

Training uses cross-entropy loss with teacher forcing, employing the AdamW optimizer and a learning rate of 3×10^{-5}. Early stopping based on validation loss is applied to prevent overfitting. The model learns to map from combined chart-text and question inputs to accurate answers, handling numerical reasoning, comparisons, and logical inferences.

3) Inference: During inference, detected visual features and their corresponding text annotations are combined into the input sequence along with the user's question. The T5 model generates the answer autoregressively, providing a natural language response based on the chart data.

V. Implementation

Our methodology involves three primary components: feature extraction from chart images using object detection models, table-based question answering using a T5 transformer model, and direct utilization of annotated text without OCR.

A. Feature Extraction with YOLO and Faster R-CNN

To extract meaningful visual features from chart images, we implement two state-of-the-art object detection models: YOLOv3 [20] and Faster R-CNN [19].

YOLOv3 (You Only Look Once) is a single-stage detector that divides the input image into grids and predicts bounding boxes along with class probabilities directly. Its real-time performance and high accuracy make it suitable for detecting chart elements such as bars, lines, legends, and axis ticks efficiently. The model processes the entire image in one forward pass, enabling rapid feature extraction.

Faster R-CNN, on the other hand, is a two-stage detector that first generates region proposals using a Region Proposal Network (RPN) and then classifies these proposals. While computationally more intensive, Faster R-CNN typically offers higher accuracy in localizing objects, which helps in precisely identifying intricate chart components that may be visually complex or overlapping.

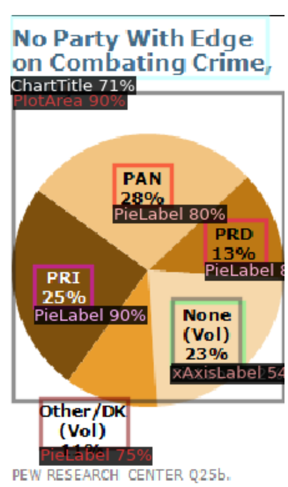

Fig. 3. Implementation of Faster R-CNN

Both models are trained or fine-tuned on annotated chart datasets where bounding boxes correspond to visual elements. Importantly, our pipeline does not rely on OCR; instead, textual information such as labels and legends is directly taken from the provided annotations associated with the dataset. This approach avoids potential OCR errors and maintains high fidelity of textual data.

B. Question Answering with T5 Transformer

For answering natural language questions based on extracted tabular data, we leverage the T5 (Text-to-Text Transfer Transformer) model [21]. T5 is a versatile sequence-to-sequence transformer pretrained on a diverse corpus, capable of handling a variety of NLP tasks by converting them into a text generation problem.

In our setup, the input to T5 is a concatenation of the question and a serialized representation of the extracted table derived from chart features and annotations. The model generates answers in natural language, performing reasoning over both the numerical and textual data. This setup enables handling complex queries including arithmetic computations, comparisons, and logical inferences.

The T5 model is fine-tuned on a question-answering dataset aligned with chart data, optimizing it for the domain-specific reasoning challenges present in visual question answering over charts.

C. Integration and Workflow

The overall workflow begins with processing a chart image through YOLOv3 and Faster R-CNN to detect and localize visual elements. Corresponding textual information is directly retrieved from annotations linked to detected elements, bypassing OCR steps. The extracted structured data is then converted into a serialized table format.

This table, combined with a user query, forms the input to the fine-tuned T5 model, which outputs the final answer. This pipeline thus integrates robust visual feature extraction with powerful language-based reasoning, enabling effective question answering on complex chart data.

However, due to the large size of the dataset and the substantial computational resources and execution time required, we were unable to fully integrate the entire pipeline into an end-to-end system. Despite this, each component of the pipeline—feature extraction with YOLOv3 and Faster R-CNN, annotation-based text retrieval, table serialization, and T5-based question answering—is implemented and ready for deployment when sufficient resources are available.

VI. Challenges

Despite building a modular and functional chart-based Visual Question Answering (VQA) system, several challenges were encountered during development and testing:

1) **Resource Constraints:** Due to the large size of the dataset and the heavy computational demands of deep learning models like T5, Faster R-CNN, and YOLOv3, we were unable to deploy a complete end-to-end pipeline within the available infrastructure. Training and inference for each module were performed separately, as full integration would require higher GPU memory and prolonged execution time, which were not feasible within our limits.
2) **Axis Label Detection in Faster R-CNN:** One of the practical limitations we encountered was with Faster R-CNN's ability to detect small or closely spaced elements, particularly the x-axis labels near the origin of charts. These elements were either partially detected or completely missed due to resolution and scale sensitivity. Improving this would require more refined bounding box annotations or anchor box tuning, which adds additional complexity.
3) **Absence of OCR:** Although our pipeline avoids the use of OCR by directly using annotation metadata, this assumes that the annotation data is perfectly aligned with the visual content. In real-world scenarios or noisy datasets, discrepancies between the visual representation and the annotation metadata may hinder accuracy.
4) **Visual Diversity in Charts:** The diversity in chart designs (e.g., variations in color schemes, overlapping legends, rotated labels, different font styles) presented difficulty for object detection models. While YOLOv3 performed well for general components, charts with cluttered or overlapping elements degraded performance.

Overall, while each individual module performs effectively in isolation, integrating them seamlessly and robustly into a complete chart VQA system remains an open engineering and research challenge due to both computational and modeling limitations.

VII. Experiments and Results

To evaluate the effectiveness of our modular Visual Question Answering (VQA) system for charts, we conducted experiments on three key components: visual element detection (YOLOv3 and Faster R-CNN), table-based QA using a fine-tuned T5 model, and the manual evaluation of partial pipeline outputs.

A. Object Detection Results

We tested YOLOv3 and Faster R-CNN for detecting chart components such as bars, legends, axis labels, and titles using a subset of annotated chart images. The models were trained and validated using bounding box annotations from the dataset.

While YOLOv3 achieved higher mean Average Precision (mAP), Faster R-CNN showed better precision on small objects but struggled with recall, especially for closely spaced or small axis labels.

B. Table-to-Text Question Answering (T5)

We fine-tuned a T5-base model using a filtered subset of question-answer pairs aligned with chart-derived tables. The training was conducted on Google Colab Pro , using a batch size of 8 and learning rate of 3e-4 for 3 epochs.

The model demonstrated strong performance on table-based questions, especially in numerical and comparative reasoning tasks.

We evaluated the performance of the T5 model on a large set of chart-based question-answer pairs. The model achieved an accuracy of **65.40%** on the test set, demonstrating promising results in handling both numerical and logical queries.

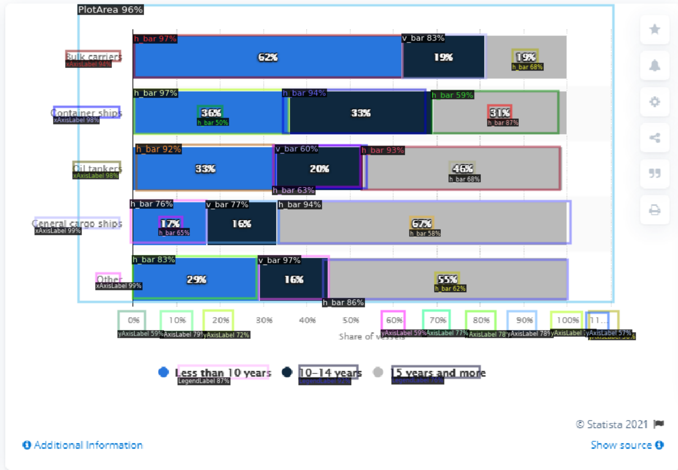

Fig. 4. Object detection

TABLE IV
SAMPLE OUTPUTS FROM T5 MODEL FOR CHART QUESTION ANSWERING

Question	Predicted Answer
In the year 2007, what is the difference in values between the first and last bars?	1.07
Are the values on the major ticks of the x-axis increasing?	No
What does the 2nd bar from the right represent?	Panama
What is the difference between the bars for 2001 and 2004?	5.4
What does the 1st bar from the bottom represent?	1990.0
How many groups of bars are there?	4.0
How many bars are there on the 4th tick of the x-axis?	3.0
What is the label of the 3rd group of bars?	2005.0
Across all countries, what is the minimum value?	3.71
How many years are there in the graph?	4.0

C. *Pipeline Integration (Qualitative Results)*

Due to computational resource limitations, we were unable to integrate all modules into a fully automated end-to-end pipeline. Instead, we manually connected outputs across modules for a small batch of test samples. Visual components were detected using YOLOv3/Faster R-CNN, annotations were parsed into structured tables, and the T5 model was used to answer user queries.

Qualitatively, the model was able to generate accurate answers when chart components were well-localized and annotations correctly matched. However, failure cases occurred when small x-axis labels were missed or bounding boxes overlapped, highlighting the need for refined detection and table conversion.

VIII. CONCLUSION AND FUTURE WORK

In this work, we presented a modular Visual Question Answering (VQA) system designed specifically for interpreting chart images. Our approach integrates object detection models (YOLOv3 and Faster R-CNN) for identifying visual elements, with a fine-tuned T5 language model for answering natural language questions based on structured chart data. Unlike traditional approaches, we bypass OCR by directly utilizing annotation data, streamlining the extraction of chart-specific text elements.

Experimental results demonstrate that YOLOv3 provides robust and fast object detection, while Faster R-CNN offers higher precision on small elements but with greater computational cost. The T5 model shows promising performance in answering a variety of table-based questions . Although we were unable to deploy a fully integrated pipeline due to resource and execution time constraints, individual components were tested and validated successfully.

Future Work

For future development, we aim to:

- Fully integrate all modules into a unified pipeline to allow end-to-end question answering from raw chart images.
- Optimize object detection to better capture small or overlapping elements such as axis labels and legends.
- Incorporate OCR-based fallback mechanisms where annotations are missing or incomplete.
- Extend support to additional chart types and more complex question types (e.g., visual comparison, trend analysis).
- Experiment with lightweight models or cloud deployment to address hardware constraints and support large-scale inference.

By refining each module and enabling full automation, our system holds promise for robust and scalable chart understanding and question answering across a wide range of visual formats.

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